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A CUMULATIVE FUSION

Based on [18], suppose we have two opinions $\overline{W}_X^A = \{b_X^A, d_X^A, u_X^A, a_X^A\}$ and $\overline{W}_X^B = \{b_X^B, d_X^B, u_X^B, a_X^B\}$. The cumulative fusion, i.e., the combined opinion, $\overline{W}_X^{A \odot B}$ of these two opinions is calculated as follows, For $u_X^A \neq 0$ or $u_X^B \neq 0$:

$$\begin{cases} b_X^{A \odot B} = \frac{b_X^A u_X^B + b_X^B u_X^A}{u_X^A + u_X^B - u_X^A u_X^B}, \\ d_X^{A \odot B} = \frac{d_X^A u_X^B + d_X^B u_X^A}{u_X^A + u_X^B - u_X^A u_X^B}, \\ u_X^{A \odot B} = \frac{u_X^A u_X^B}{u_X^A + u_X^B - u_X^A u_X^B}, \\ a_X^{A \odot B} = \frac{a_X^A u_X^B + a_X^B u_X^A - (a_X^A + a_X^B) u_X^A u_X^B}{u_X^A + u_X^B - 2u_X^A u_X^B} \text{ if } u_X^A \neq 1 \text{ and } u_X^B \neq 1, \\ a_X^{A \odot B} = \frac{a_X^A + a_X^B}{2} \text{ if } u_X^A = u_X^B = 1. \end{cases} \quad (12)$$

For $u_X^A = u_X^B = 0$:

$$\begin{cases} b_X^{A \odot B} = \gamma_X^A b_X^A + \gamma_X^B b_X^B, \\ d_X^{A \odot B} = \gamma_X^A d_X^A + \gamma_X^B d_X^B, \\ u_X^{A \odot B} = 0, \\ a_X^{A \odot B} = \gamma_X^A a_X^A + \gamma_X^B a_X^B, \end{cases} \quad (13)$$

where $\gamma_X^A = \lim_{u_X^A \rightarrow 0, u_X^B \rightarrow 0} \frac{u_X^B}{u_X^A + u_X^B}$, and $\gamma_X^B = \lim_{u_X^A \rightarrow 0, u_X^B \rightarrow 0} \frac{u_X^A}{u_X^A + u_X^B}$.

B DISCOUNTING OPERATOR

Based on [18], suppose we have two opinions $\overline{W}_B^A = \{b_B^A, d_B^A, u_B^A, a_B^A\}$ and $\overline{W}_X^B = \{b_X^B, d_X^B, u_X^B, a_X^B\}$. The discounting operator combines A 's trust in Δ , and B 's opinion about X . Then to derive A 's opinion about X , the discounting operator is used:

$$\overline{W}_X^{[A:B]} = \overline{W}_B^A \otimes \overline{W}_X^B \cdot \begin{cases} b_X^{[A:B]} = p_B^A b_X^B, \\ d_X^{[A:B]} = p_B^A d_X^B, \\ u_X^{[A:B]} = 1 - p_B^A b_X^B - p_B^A d_X^B, \\ a_X^{[A:B]} = a_X^B, \end{cases} \quad (14)$$

where $p_B^A = b_B^A + u_B^A a_B^A$ is the trustworthiness of B from A 's aspect.

Lemma B.1. If $\mathcal{A}$ fully trusts Δ on evidence collection and opinion/trust evaluation, i.e., $b_\Delta^{\mathcal{A}} = 1$, the discounting equation reads: $\overline{W}_X^{[\mathcal{A}:\Delta]} = \overline{W}_\Delta^{\mathcal{A}} \otimes \overline{W}_X^\Delta = \overline{W}_X^\Delta$.

C AVERAGING FUSION

Based on [18], suppose we have two opinions $\overline{W}_X^A = \{b_X^A, d_X^A, u_X^A, a_X^A\}$ and $\overline{W}_X^B = \{b_X^B, d_X^B, u_X^B, a_X^B\}$, which are observed in the same time period. The average fusion, i.e., the averaged opinion, $\overline{W}_X^{A \oplus B}$ of

these two opinions is calculated as follows, For $u_X^A \neq 0$ or $u_X^B \neq 0$:

$$\begin{cases} b_X^{A \oplus B} = \frac{b_X^A u_X^B + b_X^B u_X^A}{u_X^A + u_X^B}, \\ d_X^{A \oplus B} = \frac{d_X^A u_X^B + d_X^B u_X^A}{u_X^A + u_X^B}, \\ u_X^{A \oplus B} = \frac{2u_X^A u_X^B}{u_X^A + u_X^B}, \\ a_X^{A \oplus B} = \frac{a_X^A + a_X^B}{2}. \end{cases} \quad (15)$$

For $u_X^A = u_X^B = 0$, averaging fusion operates the same as cumulative fusion as in Eq. 13.

D EXPERIMENT SETUP AND ADDITIONAL RESULTS IN AIM EXPERIMENTS

D.1 CACC Attacker Model Setup

In single-directional trust model experiments, we complete 6 continuous trips and in each trip, *att* and attacking time period are different. In trip 1 and 2, vehicle 1, 2, 3 are attackers; in trip 3 and 4, vehicle 3, 6 are attackers; in trip 5, vehicle 9 is attacker and in trip 6, vehicle 3 is attacker again. In bi-directional model experiments, we complete 3 sets of trips and each contains 2 continuous trips. *att* = [1, 2, 3] in all trips and the attacking time periods are similar as well. We set attackers' trust histories in these 3 sets to be {0, 0, 1, 0.5}, {0, 0.5, 0.5, 0.5}, and {0, 0.99, 0.01, 0.5}, to represent no, moderate, and bad history.

D.2 Hyper Parameters in AIM-Trust and AIM-RL

b_{th} is a hyper parameter indicating a reasonable buffer size. We aim to find appropriate buffer sizes to decrease the collision rate and increase the throughput of the intersection. In our experiment, we let b_{th} take values 10, 12, 12, 20, 20 when the untrusted vehicle percentage is 20%, 40%, 60%, 80%, 100% respectively. The architecture of our neural network contains four layers with 4 neurons input layer, 17 neurons in output layer, and both hidden layers have 32 neurons and take ReLU activation function. The action space (buffer size) ranges from 0-16 in cases with 20%, 40%, and 60% untrusted vehicles, and 5-21 in cases with 80%, and 100% untrusted vehicles. We set the learning rate $\alpha = 0.001$, discounting factor $\gamma = 0.9$, and balancing factor $\lambda = 0.2$.

D.3 Collision-free AIM-Trust

In this experiment, we modify the reward function of AIM-Trust as as shown in Eq. 16, and let RL agent choose buffer size in range 0 to 26. AIM-Trust learns to choose larger buffer sizes to achieve collision-free in training and on average, the resulting buffer sizes in cases with 20% to 100% untrustworthy vehicles are [14.4, 14, 14.4, 20, 24]. In comparison, with same reward function, AIM-RL also learns to avoid collision completely, but with higher buffer sizes [14, 15, 16, 20, 26] which lead to lower throughput.

$$r_t^i = \begin{cases} 1, & \text{if no collision,} \\ -40, & \text{if collision happens.} \end{cases} \quad (16)$$

D.4 RL Formulation of TLC-RL

D.4.1 State Space and Action Space of TLC-RL. Similarly to the previous RL formulation in AIM, we model a four-way intersection. The states are defined as $\mathbf{s}_t = (position_t^1, speed_t^1, \dots, position_t^n, speed_t^n)^T$, where n represents the number of vehicles appeared in the map $\mathcal{M}$ at time t . The action at time t is defined as $\mathbf{a}_t = (a_t^{north}, a_t^{south}, a_t^{east}, a_t^{west})^T$, where a_t^i is the traffic light status of direction i . Note that each direction has three lanes (turn left, go straight, and turn right) and the traffic signal is the same for lanes in the same direction.

D.4.2 Reward Function of TLC-RL. To maximize the throughput and efficiency of intersections, we reward the RL agent if the cumulative waiting time of all vehicles waiting before an intersection $\mathcal{I}$ decreases compared to the previous time step, otherwise we penalize the RL agent. Therefore, the reward function is defined as follows: $r_t = W_t - W_{t+1}$, where $W_t = \sum_{i=1}^n w_t^i$ represents the cumulative waiting time of all vehicles at time t .

D.5 Hyper Parameters in TLC-Trust and TLC-RL

We keep RL formulations and training procedures of TLC-RL and TLC-Trust the same (except that TLC-Trust has trust in state space) to ensure fair comparison. The convolutional neural network architecture of our deep Q-network is as follows: IN-Conv32-Conv64-Conv128-MaxPooling-FC64-OUT, where the IN and OUT represent input and output layer. ReLU activation function is placed in every convolutional and fully-connected layer. We set the learning rate $\alpha = 0.001$, discounting factor $\gamma = 0.99$,

E DEMONSTRATION VIDEOS AND CODES

In Supplementary Materials folder, we include demonstration videos of AIM-1, AIM-RL, AIM-Trust (collision-free), TLC-Fix, TLC-RL, TLC-Trust along with their codes and ‘readme’ files.